

Influence of superimposed bedforms and flow unsteadiness on formation of cross strata in dunes and unit bars

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Abstract

Formation of angle-of-repose cross strata on the lee side of dunes and unit bars is dependent on three main grain-sorting mechanisms: (1) presorting of sediment that arrives at the lee side of the bedform, related to superimposed bedforms and longer-term variations in water flow and sediment transport; (2) sorting due to differential deposition of sediment on the lee side and associated grain flows, and; (3) movement of sediment on the lee side by the water currents in the lee side flow-separation zone. Although most emphasis has been put on mechanism (2), recent field and experimental studies of the dynamics of sandy and gravelly dunes and unit bars show that mechanism (1) is at least as important.

Bedforms superimposed on the backs of dunes and unit bars are ubiquitous, and are composed of size sorted sediment. These superimposed bedforms travel faster than the host bedform and overtake it. If the overtaking bedform has a relatively large height, the lee side of the host bedform becomes reduced in slope and decelerates as the superimposed bedforms overtake. This results in a “reactivation surface” that is commonly lined with relatively fine-grained sediment. However, if an overtaking superimposed bedforms has a relatively small height (e.g., a ripple or bedload sheet), the lee side of the host bedform remains at the angle-of-repose. The sediment within the overtaking bedform forms a relatively thick cross stratum on the lee side of the host bedform. Arrival of the trough of the next superimposed bedform at the crest of the host bedform results in deposition of a relatively thin stratum. The thickness of the cross stratum couplet formed in this way is related to the height and length of the superimposed bedform and the height of the host bedform, which are related to flow depth and dimensionless bed shear stress on the back of the host bedform. The grain size of the sediment in the superimposed bedforms varies with their height, and this is reflected in the grain size of successive cross strata. The formation of cross-stratified open-framework gravel is associated with size sorting of sand and gravel in superimposed bedforms, and further size sorting associated with flow-separation on the lee side of large host bedforms.

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1. Introduction

Cross-stratification formed by dunes and unit bars is the most common sedimentary structure in river-channel deposits, and is common in many other depositional environments. Ancient cross-stratified deposits are

widely used to interpret conditions of flow and sediment transport, such as water depth, flow velocity magnitude and direction during deposition (review in [Bridge, 2003](#)). Cross-strata are defined primarily by spatial variation in grain size, which in turn affects spatial variation in permeability, and the movement of fluids through cross-stratified deposits. Grain size sorting in cross strata is determined by three main factors ([Fig. 1](#)): (1) presorted sediment that arrives at the brink point, related to

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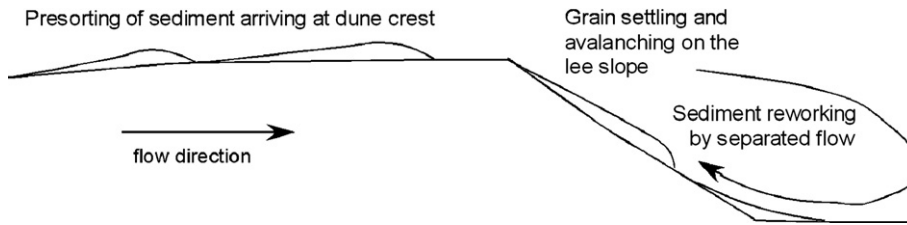


Fig. 1. Controls on the grain size sorting in cross strata formed by dunes.

superimposed bedforms and longer-term unsteadiness in water flow and sediment transport; (2) sorting due to differential deposition of sediment on the lee slope and associated grain flows, and; (3) movement of sediment on the lee side by water currents in the lee side flow-separation zone. Factor (2) is commonly taken as dominant. However, recent experimental and field studies indicate that the role of bedform superimposition (e.g., ripples or bedload sheets on dunes; ripples or dunes on unit bars) in the formation of cross-stratification is much more important than commonly appreciated.

This paper will start with a short outline of our understanding of the formation of cross strata, identify gaps in our knowledge, and address the common misconception that grain flows cause the formation of most, if not all, cross-stratification. Then, new experimental and field data will be presented that demonstrate the importance of presorting of sediment, specifically superimposition of bedforms, in determining the grain size sorting in cross strata. Finally, further research on this topic is suggested in order to allow for more accurate prediction of spatial variation in porosity and permeability (of concern to hydrogeologists and petroleum geologists), and to improve quantitative interpretation of ancient flow and sediment transport conditions.

2. Controls on the grain size sorting in cross strata formed by dunes and unit bars

2.1. Presorting of sediment due to superimposed bedforms and longer-term unsteadiness in water flow and sediment transport

2.1.1. Long-term flow unsteadiness

Changes in the rate and grain size of sediment supply to the crests of bedforms may be related to relatively long-term flow unsteadiness associated with weather-related floods, bank-slumps, and cyclical variations in discharge such as caused by snowmelt and freezing in cold regions, or tides. The grain size of bedload is expected to decrease as flow stage and bed shear stress decrease (review in [Bridge, 2003](#)). It has been suggested that relatively coarse-grained cross strata are deposited at high flow stage, and that fine-grained cross strata are deposited at low flow stage ([Smith, 1972, 1974; Steel and Thompson, 1983; Smith, 1985](#)). Alternations of sandy cross strata and muddy drapes in tidal dunes (called tidal bundles: [Fig. 2](#)) are generally taken to be formed by alternating periods of tidal current flow and tidal slack water (e.g. [Visser, 1980; Allen, 1982; Nio et al., 1983](#)). Systematic variation in the thickness of tidal



Fig. 2. Tidal bundles in an excavation in the Oosterschelde (courtesy of Dr. Van den Berg). Height is approximately 2 m.

bundles has been related to variations in tidal range associated with neap-spring cycles and with the diurnal inequality of the tide. However, the exact nature of the effects of long-term flow unsteadiness on sorting in cross strata has never been examined systematically using flume experiments.

2.1.2. Superimposed bedforms

Related to flow unsteadiness is temporal change in the geometry of dunes, and the likelihood that dunes of different sizes will be superimposed (reviews in Allen, 1982; Bridge, 2003). However, experimental and field data show that ripples and bedload sheets occur on the backs of dunes, and bedload sheets, ripples and dunes occur on the backs of unit bars, under both steady and unsteady flow conditions (Allen, 1982; Bridge, 2003; Venditti et al., 2005; Lunt and Bridge, 2007: Fig. 3). Furthermore, it has been recognized that many bedforms grow by amalgamation of smaller bedforms, and that small, embryonic bedforms are continually forming on the backs of larger ones (Mantz, 1978; Allen, 1982; Raudkivi and Witte, 1990; Ditchfield and Best, 1992;

Gabel, 1993; Coleman and Melville, 1994, 1996; Raudkivi, 1997; Leclair, 2002). The influence of these ubiquitous superimposed bedforms on the nature of cross strata has never been investigated systematically.

The influence of superimposed bedforms on the development of cross strata in unit bars was investigated early on by Hooke (1968), Smith (1972), and Rust (1984). Hooke (1968) and Smith (1972) observed how granules accumulate in the troughs of sandy superimposed bedforms, such that granule cross strata in unit bars is associated with the passing of the troughs of the superimposed bedforms (Fig. 4). Rust (1984) argued that open-framework gravel occurred within isolated dunes superimposed on unit bars, whereas bedload sediment in the absence of superimposed dunes was a mixture of sand and gravel (Fig. 4). Therefore, cross strata formed of open-framework gravel, bounded by cross strata formed of sandy gravel, were attributed to deposition from superimposed dunes alternating with deposition from non-dune regions. Experiments with sandy gravels by Lunt and Bridge (2007) indicated that bedload sheets and dunes superimposed on unit bars are composed mainly of open-framework gravels,



Fig. 3. Bedforms superimposed on dunes and unit bars. (A) Bedload sheets on the back of sand dune. Dark areas are troughs with relatively fine-grained sediment. Flume is 0.6 m wide. (B) Ripples on sand dunes. Flume is 0.6 m wide. (C) Ripples on a unit bar in sand. Flume is 0.6 m wide. (D) Dunes and people on crest of a sandy unit bar, Congaree River. Trough of unit bar is to left.

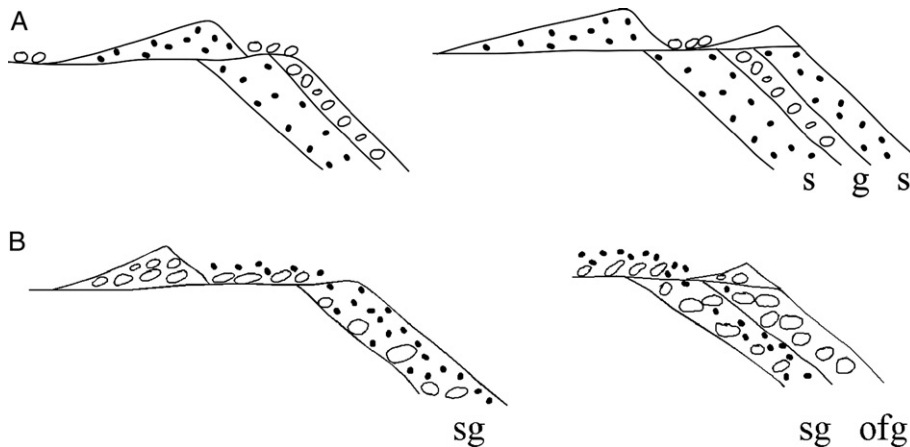


Fig. 4. Grain size sorting in unit bars resulting from size sorting in superimposed dunes, according to (A) Hooke (1968) and Smith (1972), and (B) Rust (1984). s is sand, g is gravel, sg is sandy gravel and ofg is open-framework gravel.

as suggested by Rust (1984). The highest superimposed bedforms have the best developed open-framework gravels, but the smaller superimposed bedforms are composed of sandy gravel. Therefore, cross strata in unit bars composed of alternations of open-framework gravel and sandy gravel (Fig. 5) are related to variation in the height of superimposed bedforms (c.f. Rust, 1984). These experiments also indicate that a veneer of suspended fine-grained sand is deposited in the trough of bedload sheets and dunes, and partly infiltrates the coarse-grained sediment in this region. Such co-accumulation of small and large grains in bedform troughs results in highly bimodal sediments. As a superimposed bedform trough arrives at the brink point of a host bedform, fine-grained suspended load and bimodal trough sediment are deposited on the lee side of the host. As the superimposed bedform arrives, dominantly coarse-grained sediment is deposited on the lee side of the host (Fig. 6).

If superimposed bedforms are high relative to the host bedform ($H_s/H > 0.2$ where H_s and H are heights of superimposed and host bedforms, respectively), the lee side slope of the host bedform becomes reduced as the superimposed bedforms approaches its crest, due to erosion associated with the flow-separation zone of the superimposed bedforms. This overtaking process produces a low-angle “reactivation” surface rather than an angle-of-repose cross stratum (Fig. 6), and the internal structure of the host bedform is composed of cross-sets made by the superimposed bedforms with set boundaries dipping downstream at less than the angle-of-repose (Fig. 6). However, an identical structure (called compound cross-stratification by Allen, 1982) can be formed by superimposed bedforms migrating over a larger host bedform that does not have a lee face at the angle-of-repose, because it is either developing or dying out (Fig. 7). This is



Fig. 5. Experimental data from Lunt and Bridge (2007) showing cross strata of varying thickness and grain size within a unit bar, associated with deposition from superimposed bedforms of varying height. Peel is 0.5 m across and 0.2 m high.

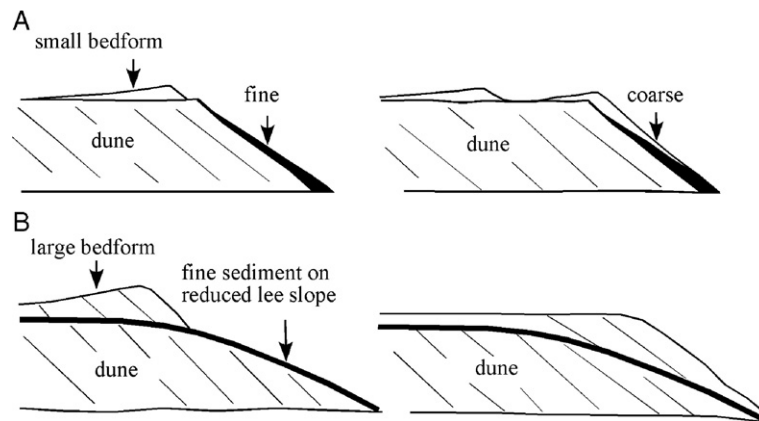


Fig. 6. (A) Formation of cross strata by migration of small bedforms superimposed on a dune back as observed in recent experiments (see also Figs. 4 and 5). (B) Formation of a “reactivation surface” by migration of a large bedform over a dune.

commonly produced as ripples or dunes migrate over developing unit bars (e.g., Jackson, 1976; Levey, 1978; Bridge et al., 1986; Best et al., 2003; Lunt et al., 2004).

2.2. Sorting due to differential deposition of sediment on the lee side and associated grain flows

Bedload sediment and near-bed suspended sediment that arrives at the brink point of the bedform is deposited on its lee side at a distance downstream from the brink point that is controlled by: (1) grain velocity; (2) height above the bed; (3) grain settling velocity, and; (4) turbulence in the separated flow (Fig. 8; Jopling, 1965; Allen, 1963, 1968, 1982; Hunter, 1985b). Grains travel further downstream from the brink point as their velocity and height above the bed increase, and as their settling velocity (hence grain size) decreases. Therefore, the rate of deposition and the grain size of deposited sediment decrease down the lee slope, resulting in building of a down slope thinning wedge of sediment that fines down slope (i.e. tangentially).

The slope of the outer edge of the wedge of sediment that builds up on the lee side increases to the angle of initial yield, at which point a grain flow occurs and comes to rest at the so-called residual angle. The angle of initial yield is commonly around 35°, and up to 10° larger than the residual angle after grain flow. In reality, the angle of yield may not reach its maximum value prior to grain flow, because turbulence in the flow-separation zone can cause the sediment wedge to fail prematurely. The lee side wedge builds quickly if sediment transport rate is high, and if most of the sediment load is near the bed. Thus, grain flow frequency increases with sediment transport rate, provided that most material is deposited on the upper lee slope. The size (cross-sectional area) of

the wedge of sediment that eventually flows down the lee side depends on the spatial variation of deposition rate on the lee slope. For coarse-grained bedload sediment

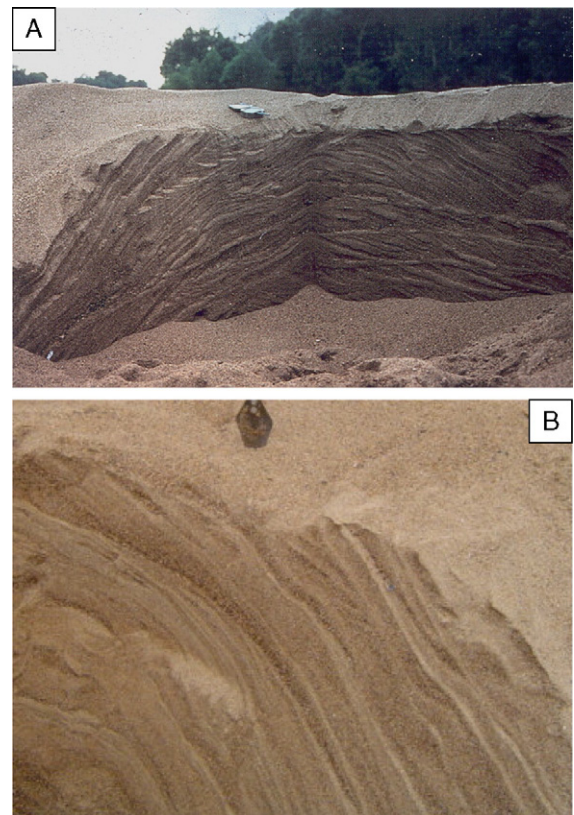


Fig. 7. (A) Compound cross strata formed by migration of dunes over unit bar, Congaree River. Angle-of-repose cross strata associated with the avalanche face of the unit bar only occur at the downstream edge of the unit bar. (B) Close-up of outer edge of unit bar deposits showing grain size sorting due to dune migration. The finest layers are due to passing of the dune troughs. Note internal lamination and distinct reactivation surface below trowel.

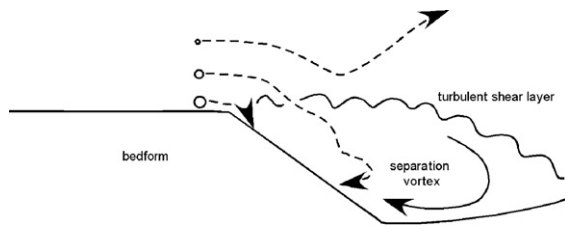


Fig. 8. Idealized paths of grain fall on bedform lee side for grains of different size and distance from the bed.

moving relatively slowly, most sediment is deposited close to the brink point, and the failing wedge of sediment is relatively small. The lack of settling of suspended sediment on the lower lee side results in an angular contact between the lee side slope and the bedform trough (Fig. 9). For fine-grained near-bed sediment moving relatively fast, much of the sediment is transported well past the brink point and even into the trough, and the wedge would be larger. The build up of fine-grained sediment at the base of the lee side reduces the angle of the lower slope (tangential bases: Fig. 9). Small sediment wedges at the point of failure result in small grain flows that may not involve enough material to reach the base of the lee slope. However, subsequent addition of sediment from other small grain flows can produce a large composite wedge that can reach the base of the lee slope (Hétu et al., 1995). There are no realistic models for the size of the avalanching wedges relative to bedform size, nor for the thickness of grain flow deposits. Grain size sorting within grain flows originating from composite wedges is also poorly known.

Differential rates of deposition of different sediment sizes on the angle-of-repose lee side of bedforms, and associated grain flows, are commonly taken as the most important controls of grain size sorting in cross strata (Allen, 1963; Jopling, 1965; Allen, 1968, 1970, 1982; Hunter, 1985a,b; Hunter and Kocurek, 1986; Carling and Glaister, 1987; Makse, 1997; Makse et al., 1997a,b; Koepppe et al., 1997; Kleinhans, 2004, 2005). Grain size sorting during discrete grain flows has been related to

kinetic sieving (Middleton and Southard, 1984) and to dispersive grain stress (Bagnold, 1954; Sallenger, 1979), although the latter mechanism has been substantially discredited (Legros, 2002). With kinetic sieving, small grains settle between the spaces created between the larger grains during grain shearing and dilation. The large grains move towards the faster-moving outer edges of the grain flow, where they can move down slope more quickly than smaller grains. The largest grains on top of the grain flow roll down the slope relatively easily because they have small friction angles. Therefore, the grain flow deposit becomes coarser grained down slope and towards its outer edge (Fig. 9). The bedform must be high enough (lee side must be long enough) for this process to be effective. At relatively low sediment transport rates, grain flows are intermittent, and it may be possible to distinguish relatively coarse-grained grain flow strata from relatively fine-grained grain-fall strata (Jopling, 1965; Hunter, 1985a). At high sediment transport rate, when grain flows are very frequent, much fine-grained suspended sediment is incorporated during avalanching and also deposited on the lower lee slope (Allen, 1970, 1982; Hunter, 1985a,b; Carling and Glaister, 1987; Kleinhans, 2004). This may counteract the sorting caused by kinetic sieving, such that grain size sorting may not be obvious.

Avalanching of sediments with a large range of grain size, such as sandy gravels, may lead to complete segregation of different grain sizes along the angle-of-repose slope, and to absence of cross strata (e.g. Koepppe et al., 1997; Makse, 1997; Makse et al., 1997a,b, 1998; Fig. 10). This is due to the large differences in pivoting (friction) angles between the different grain sizes in a mixture. The largest grains moving over smaller grains have the smallest pivoting angles and readily move to the base of the lee slope. This means that coarse-grained cross strata in many sandy gravels cannot be related to sorting in grain flows and instead must be related to superimposed bedforms or longer-term flow unsteadiness. Unfortunately, most of the experimental studies (cited above) of the nature of sediment deposition and

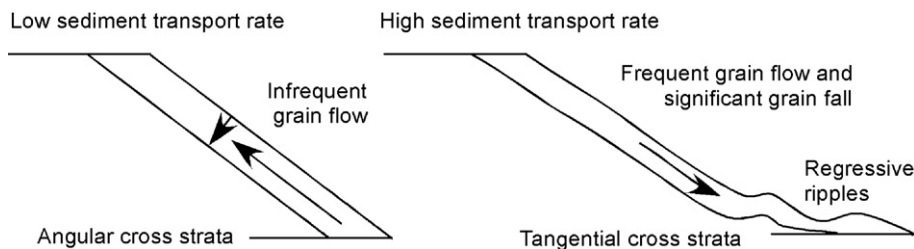


Fig. 9. Grain size sorting by grain flow and grain fall on bedform lee side, and geometry of cross strata, for low and high sediment transport rate. Arrows point to direction of decreasing grain size.



Fig. 10. Experimental grain flow of sandy gravel produces grain size separation but no cross strata.

grain flow on lee slopes did not involve serious consideration of variations in rate and size of sediment arriving at the brink point. It is clear, however, that grain flow cannot explain many types of cross strata.

2.3. Movement of sediment on the lee side by the water currents in the flow-separation zone

The separated (reverse) flow over the bedform lee side can reduce the velocity of the grain flows and move sediment up the lower lee slope (in some cases as ripples: Boersma et al., 1968), causing a slope decrease there. It is readily observed in flume experiments that grain flows do not always reach the base of the lee slope. However, cross strata commonly continue into the trough (Fig. 9). This tangential cross-stratification cannot be formed by grain flows, and can only be associated with deposition of suspended sediment and/or action of the reverse-flow. These processes also need to be investigated systematically.

3. Recent experimental studies

Experiments were initiated in the Binghamton flume in order to: (1) generate cross strata within dunes and unit bars formed in sands and sandy gravels under steady and unsteady flow conditions; (2) describe the geometry, grain size sorting, porosity and permeability of cross strata formed under different conditions; (3) evaluate the controls on the formation of the cross strata, and; (4) examine the possibility of using the cross strata to interpret formative flow conditions in much more detail than possible now. A prime objective is to be able to distinguish the effects of sorting from grain flows and presorting on the back of the bedforms. In reality, grain

size sorting probably results from a combination of presorting, lee side grain flows, and reworking by the lee side flow vortex. Preliminary results are presented here for the cases of cross strata formed by ripples and bedload sheets superimposed on dunes, and ripples and dunes superimposed on unit bars. The results are compared with cross strata formed by subaqueous grain flows developed in a Hele-Shaw cell, in which presorting can be assumed negligible.

3.1. Binghamton flume and experimental procedure

The laboratory flume in the Department of Geological Sciences, Binghamton University, is 0.61 m wide, 0.4 m deep, and 7.6 m long. It recirculates water and sediment up to about 50 mm in diameter, using a combined propeller pump (for water and suspended sediment) and diaphragm pump (for water and gravelly bedload sediment). Sand and water can be recirculated using only the propeller pump. A conveyor belt at the upstream end of the flume allows introduction of sediment across the width of the flume at a constant rate, for use in experiments with aggradation. An ultrasonic depth profiler is used for measuring bed elevation to within 0.1 mm, and can easily resolve mm-high bedforms. A computerized instrument carriage can be used to move the depth profiler along the flume at a fixed rate. A Marsh-McBirney electromagnetic flow meter with a 12 mm diameter sensing head is available for measuring profiles of flow velocity.

Two sediment mixes were used in the experiments (Fig. 11): moderately well-sorted medium sand ($D_{50}=0.43$ mm), and poorly-sorted sandy gravel ($D_{50}=2$ mm). Bedforms were developed from an initially flat bed. The flume was run for at least the time it

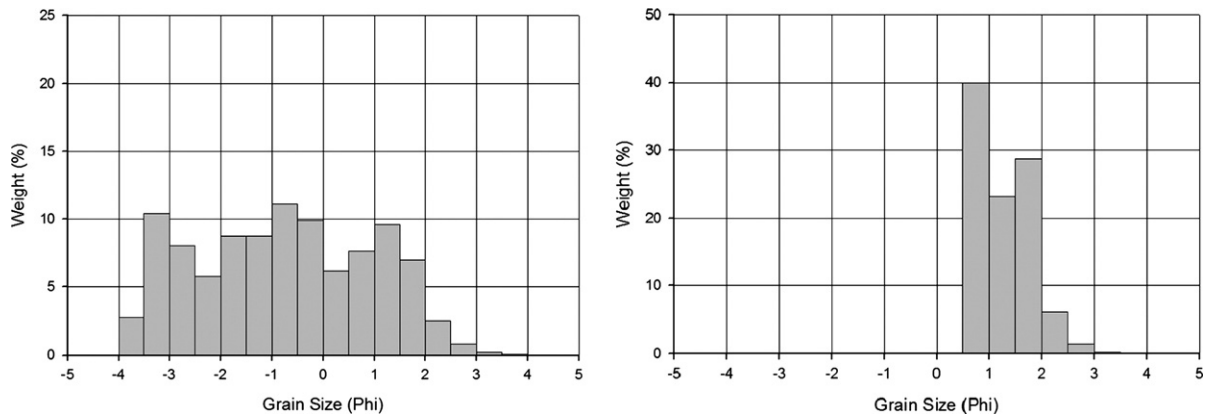


Fig. 11. Grain size distribution of the sand and sandy gravel used in the experiments.

takes one dune to migrate through the entire length of the flume before measurements were made. Unit bars were developed in the flume by overloading the upstream end of the flume with sediment. Superimposed bedforms of specific types and sizes were controlled by varying the sediment feed, flow velocity and water depth (Table 1). Water–surface elevations were measured throughout the experiments through the sidewalls along the length of the flume. Water–surface slopes in the downstream half of the flume were derived from regression analysis of repeated measurements of water–surface elevation. The experiments were recorded on videotape, which facilitated a detailed analysis of the depositional processes. Following each experiment, the flume was drained, photographs were taken of the bed and the deposits seen through the flume sidewalls, and cores were taken in various orientations. Epoxy-resin peels were made

of some of the cores, and others were used for grain size analysis and determination of permeability and porosity.

The experiments are true scale, meaning that the flow, sediment transport and bedforms are representative of natural rivers with relatively small flow depths and subcritical flow. The results could be upscaled to deeper flows insofar as the heights and lengths of dunes are proportional to flow depth, the heights of unit bars are proportional to flow depth, and their lengths are proportional to flow width. However, upscaling must also involve other parameters, such as grain size, which must be scaled up in the same ratio as flow depth.

3.2. Grain flow apparatus and experimental procedure

The Hele-Shaw cell is a metal box (0.04 m wide, 0.5 m long and 0.2 m high) with a removable side plate

Table 1
Experimental conditions used in the flume

Sediment type	Depth average velocity (m/s) ^a	Average water depth (m) ^a	Average water surface slope (–)	Froude number	Host bedform	Superimposed bedforms	Aggradation rate (mm/s)
Sandy gravel	0.47 (0.87)	0.20 (0.11)	–0.007	0.34 (0.84)	Unit bar	Bedload sheets, dunes	0.019
Sandy gravel	0.84	0.16	–0.004	0.67	Bedload sheets, dunes	–	0.0075
Sandy gravel	0.34 (–)	0.22 (0.09)	–0.004	0.23 (–)	Unit bar	Bedload sheets, dunes	0.018
Sandy gravel	0.20 (0.75)	0.20 (0.06)	–0.005	0.14 (0.98)	Unit bar	Bedload sheets, dunes	Unsteady
Sand	0.55	0.14	–0.004	0.47	Dune	Ripples	0
Sand	0.65	0.14	–0.003	0.55	Dune	Ripples, bedload sheets	0
Sand	0.75	0.14	–0.005	0.64	Dune	Bedload sheets, ripples	0
Sand	0.85	0.14	–0.006	0.73	Dune	Bedload sheets	0
Sand-test run	–	0.35 (0.10)	–	–	Unit bar	Ripples	0.02
Sand-test run	–	0.20 (0.15)	–	–	Unit bar	Ripples	0.02
Sand	0.19 (0.6)	0.25 (0.08)	–	0.13 (0.68)	Unit bar	ripples	0.12
Sand	0.19 (0.6)	0.25 (0.09)	–	0.13 (0.68)	Unit bar	Ripples, dunes with ripples	0.08

– Numbers not available or not significant.

^a First number is measured in trough of unit bar (number in brackets measured on back of unit bar).

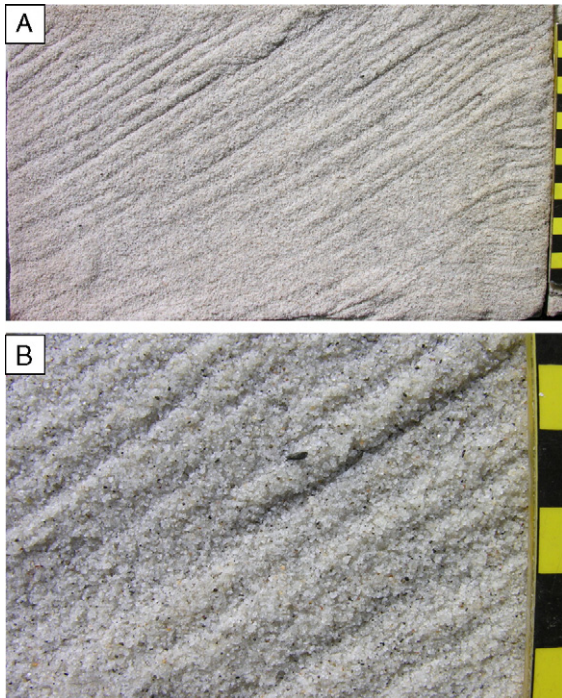


Fig. 12. (A) Peel of subaqueous grain flows in sand. Peel is 0.3 m wide and 0.18 m high. (B) Section of the upper part of the peel. Scale on the right side is in centimeters.

that allows epoxy peels to be made. Sediment was fed to the Hele-Shaw cell through a 2D funnel at one end of the box initially. Once the sediment reaches the angle-of-repose, the sediment slope builds to the top of the box by repeated grain flows. The slope is then allowed to build out along the length of the box at constant height by moving the funnel slowly across the top of the box. Subaqueous grain flows were made by placing the entire apparatus under water. Grain flow deposits of height less than the box height was made by partially filling the Hele-Shaw cell with sediment. In order for grain flow deposits to be compared with the cross strata developed in the flume, their heights were 0.07 m and 0.18 m. In addition, observations were made in a Hele-Shaw cell of similar dimensions made out of Plexiglas (after Makse et al., 1998).

4. Experimental results

4.1. Hele-Shaw cell experiments with grain flows in sand

The cross strata formed in the Hele-Shaw cell by subaqueous grain flows are picked out clearly on the peels by variations in grain size (and permeability), particularly in the upper parts of the peels (Fig. 12). A veneer of fine sand occurs at the base of a cross stratum,

and grain size generally increases upward within the cross stratum to medium sand. The true thickness of grain flow deposits (measured normal to the slope) ranges from 4.5 to 8 mm, with a mean of 6.2 mm. Grain flow deposits that are 0.18 m and 0.07 m high have cross strata with the same thickness (Table 2). The angle of the cross strata varies from 34° at the top to about 29° at the bottom. The contact between the cross strata and the bottom of the Hele-Shaw cell is angular. Grain flows that reached the bottom of the Hele-Shaw cell were built up from several smaller flows that did not contain enough sediment to reach the bottom. The flow mechanics were the same for the small and large grain flows. The relative uniformity of grain flow thickness is due to the limiting volume (weight) of sediment required to allow the flow to reach the bottom.

4.2. Flume experiments with dunes in sand

Sand dunes formed in water depth of 0.2 m and 0.15 m, with flow velocity ranging from about 0.55 m/s to 0.85 m/s (Table 1). The dunes were about 0.05 m high and 1 m long, and had superimposed ripples and bedload sheets, with heights of 0.003 m to 0.03 m and lengths of 0.1 m to 0.2 m (Fig. 3). Ripples and bedload sheets are differentiated by an arbitrary cutoff height of 3 mm based on flume observation and existing literature (Best and Bridge, 1992; Bennett and Bridge, 1995; Bridge, 1997; Bridge and Best, 1997). There is a continuum between bedload sheets and ripples because bedload sheets on the backs of dunes can amalgamate and grow into ripples. Ripples were superimposed on dunes at relatively low flow velocities (<0.65 m/s), whereas primarily bedload sheets were superimposed on dunes at higher flow velocities.

Both ripples and bedload sheets arriving at the crest of a dune affected the shape and migration rate of the dune lee side, and influenced the nature of deposition there (Fig. 13). As ripples have a large relative height (H_r/H greater than about 0.3–0.4), the lee side of the dune is

Table 2

Details of subaqueous cross strata produced in the Hele-Shaw cell, using different grain flow heights and sediments

Sediment type	Total height (m)	Average thickness (mm)	Standard deviation of thickness (mm)	Number of measurable strata
Sand	0.18	0.65	0.09	33
Sand	0.12	0.62	0.10	11
Sand	0.07	0.58	0.08	24
Sandy gravel	0.18	—	—	0

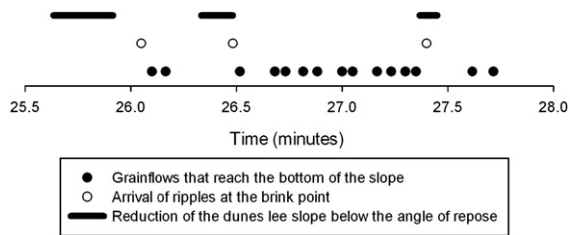


Fig. 13. Timing of the occurrence of: (1) superimposed bedforms (ripples of about 0.01 m height) arriving at the brink point of a sand dune; (2) grain flows that reach the base of the dune lee slope; and; (3) reduction of the lee side slope of the dune below the angle-of-repose (note lack of grain flows when slope angles are reduced).

commonly reduced in slope when the trough of the ripple approaches the dune crest. Bedload sheets do not tend to cause such slope reduction because of their limited height. The arrival of the body of the superimposed bedform re-establishes the angle-of-repose lee slope, and is associated with rapid advance of the dune lee slope and frequent grain flows (Fig. 13). Most grain flows were associated with the passage of the main body of superimposed bedforms. However, some grain flows were initiated by turbulent eddies impinging on the dune lee slope, and these tend to be thinner than those initiated by depositional oversteepening. The toes of grain flows were in some cases slowed down and even lifted up into the flow by the reverse-flow of the separation zone vortex. The increased dilation associated with the introduction of water into the grain flow possibly decreases the effectiveness of kinetic sieving while increasing the effect of differential settling. Grain size sorting associated with these effects requires further study.

As the superimposed ripples normally caused reduction in slope of the dune lee side, the internal structure of the dunes commonly contained compound cross strata (Fig. 14). However, angle-of-repose cross strata are formed on dune lee slopes in association with the arrival of bedload sheets. Fig. 15A and B clearly show discrete cross strata on the order of millimeters thick being formed by migration of successive bedload sheets. The cross strata are recognized by virtue of changes in grain size from fine to medium sand, as in the case of the Hele-Shaw experiments. The cross strata are tangential because of the effects of grain fall and reworking of sediment by the reversed flow. The cross strata have a maximum angle of about 32° at the top and can be traced into very-low-angle strata and regressive ripples at the base. The cross strata are distinctive from the top to the base of the set, in contrast to the cross strata formed in the Hele-Shaw cell. The average thickness of the cross strata is about 8 mm, and there is more variability in thickness than in the Hele-Shaw grain flow experiments.

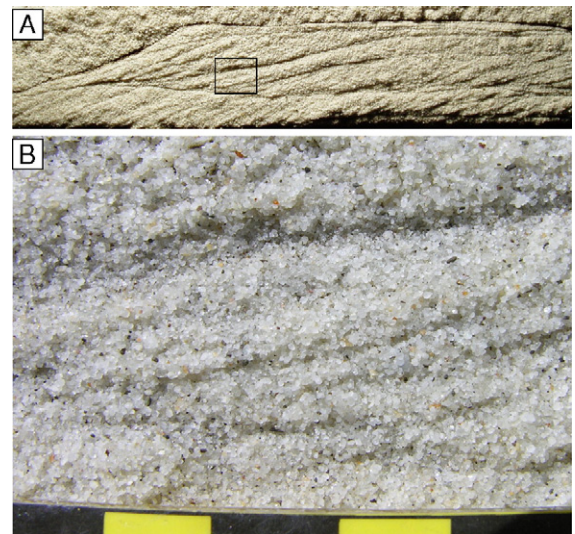


Fig. 14. (A) Peel (0.5 m wide) of deposits of sand dunes with superimposed ripples. Note the compound cross-stratification with reactivation surfaces. (B) Close-up of the same peel: the reactivation surfaces are lined with finer material.

4.3. Flume experiments with solitary unit bars in sand

Solitary unit bars in sand varied from 0.08 to 0.18 m in height (Table 1). Bedforms superimposed on sandy unit bars were ripples with heights up to 0.02 m and lengths up to 0.20 m (H_s/H of about 0.1–0.2). In view of the relatively small H_s/H , the arrival of ripples at the crest of the unit bar did not cause reduction of its lee slope. Most grain flows are associated with passage of

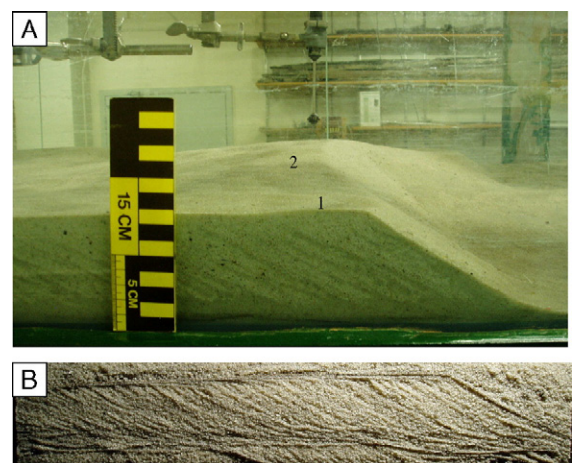


Fig. 15. (A) Low bedforms, classified as bedload sheets because of their limited height, (1,2) superimposed on dune. Bedform 1 is forming a cross stratum. Note cross strata of variable thickness that is highlighted by dark, fine-grained layers. (B) Peel (0.5 m wide) of cross strata formed in sand dune with superimposed-bedload sheets.

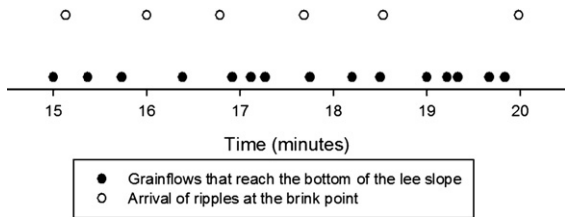


Fig. 16. Timing of arrival of ripples at the brink of a sandy unit bar, and grain flows down the lee side. Most grain flows are associated with the passage of the main body of superimposed bedforms, some are triggered by turbulence associated with their troughs.

the main body of the ripples, and several large grain flows occur as each ripple passes (Fig. 16). The relatively larger height/flow depth of unit bars compared to dunes ($\sim 2/3$ for unit bars and $\sim 1/4$ for dunes in these experiments) results in greater deceleration on lee sides of unit bars relative to dunes (flow velocity in the trough of the unit bar ~ 50 – 25% of flow velocity over its top), enhanced settling of fine-grained suspended sediment in

the unit bar trough, and formation of bottomsets that are distinctly finer-grained (very fine sand) than the cross strata (medium to fine sand) (Fig. 17). Co-flow (e.g. Fig. 3C) and reverse-flow ripples (e.g. Fig. 18B) are present in the unit bar trough depending on the flow characteristics in the trough.

The cross strata have an average thickness of 12.5 mm and can be traced into the strata of the topsets on the unit bar (Fig. 17B). The cross strata formed by migration of the unit bar are two or three times thicker than those formed by grain flows of the same height in the Hele-Shaw cell. The cross strata are recognized due to internal grain size variations from medium to fine sand. The number of cross strata preserved corresponds to the number of ripples arriving at the bar crest and not with the number of grain flows that reached the bottom of the lee slope. In a few examples, cross strata formed by migration of ripples over unit bars show internal lamination with the same thickness and grain size sorting as expected from grain flows (Fig. 7B). Some cross strata have small reactivation surfaces at the top,

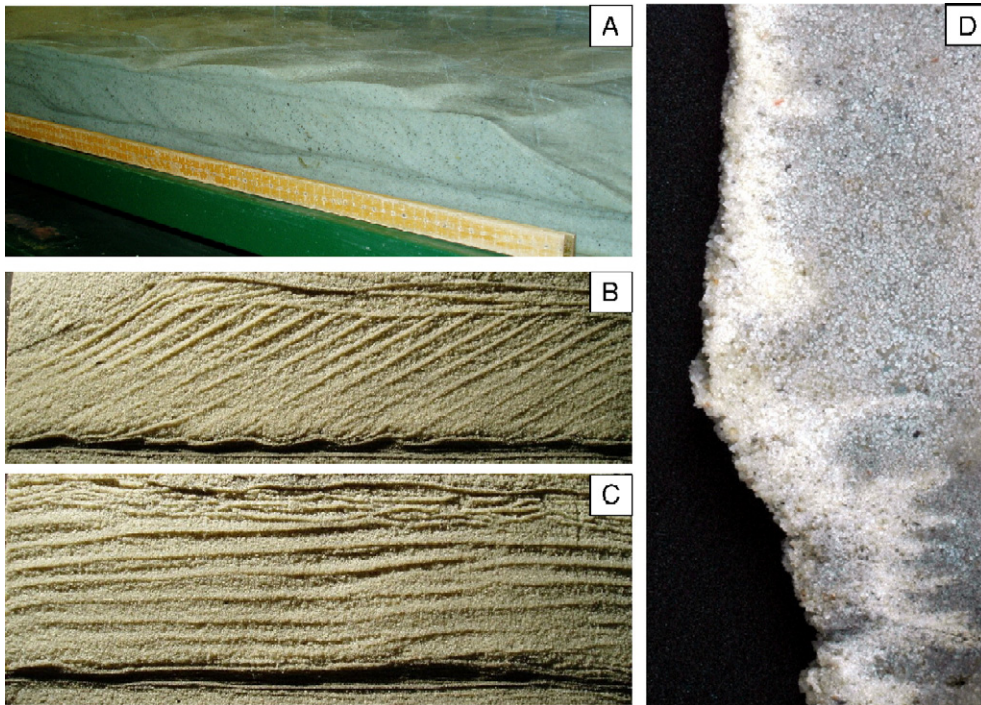


Fig. 17. (A) Sandy unit bar with superimposed ripples. Scale is a meter rule. (B) Peel (0.5 m wide) of sandy unit bar deposits with cross strata formed by superimposed ripples. Strata delineated by fine-grained drapes that stick out because of capillary effects. Note faint internal laminations caused by grain flows. (C) Peel (0.5 m wide) of across-flume view of the cross strata associated with peel B. (D) Cut section of a unit bar peel that shows the grain size variation of peel B. The dimensions of the section are 26 by 60 mm. Resin percolated from the right to the left. The light parts on the left side are partially impregnated sediment, the right side is fully impregnated sediment. The bottomsets in the lower right are composed of dark, very fine sand with internal laminations (partially impregnated). The foresets in the middle and upper right show fully impregnated medium to fine sand, indicating the higher permeability.

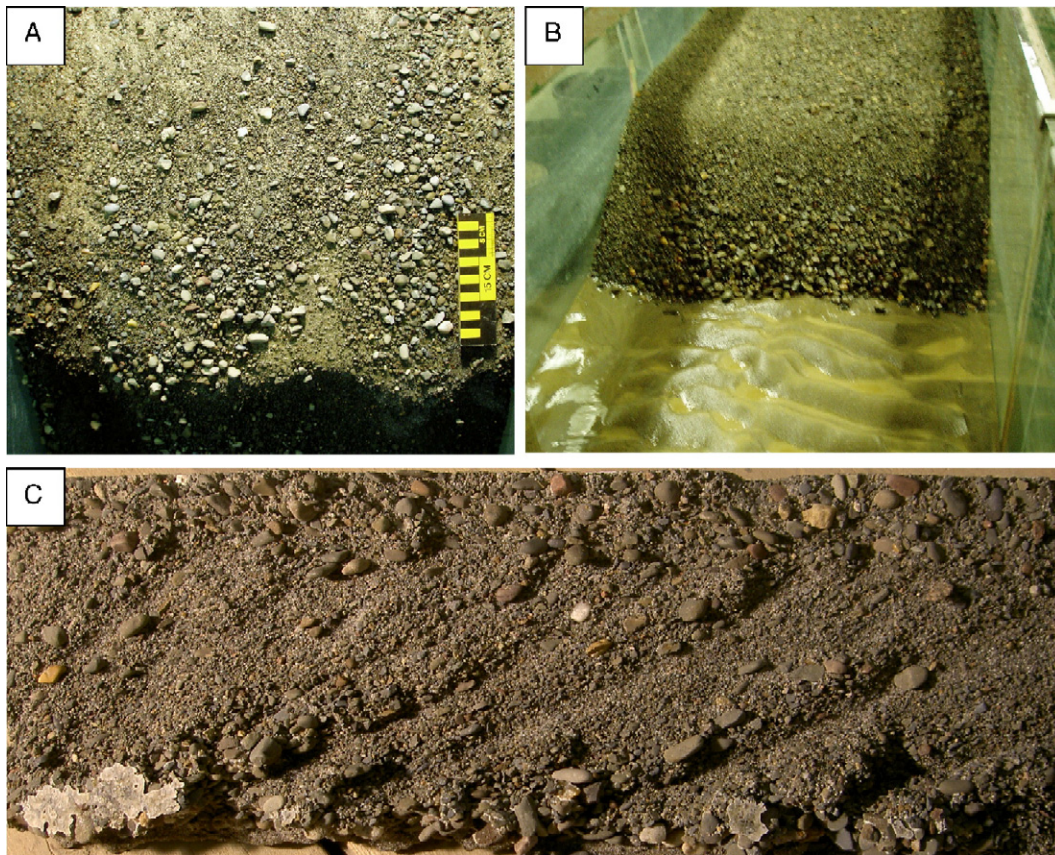


Fig. 18. (A) Unit bar in sandy gravel with superimposed-bedload sheets and dunes (B) lee side of unit bar showing down slope increase in grain size and sharp boundary with rippled fine-grained sand in the trough. Flume is 0.6 m wide. (C) Peel (0.5 m wide) of deposits of unit bar showing cross-strata of varying thickness and grain size, associated with deposition from superimposed bedforms of varying height.

which allows them to be directly associated with the passing of a superimposed bedform (Figs. 7B, 17B). The contact between the cross-set (foreset) and the bottomset is angular (Fig. 17B), indicating that grain flow rather than grain fall dominates the deposition on the lee slope.

The cross strata within the unit bar are laterally continuous and planar in the across-flume view (Fig. 17C), even though most of the superimposed ripples have sinuous crest-lines. This is because ripple crest-lines were mostly laterally continuous (see Fig. 3C). However, in some experiments (Table 1), the superimposed ripples occasionally amalgamated into dunes, which caused the formation of large reactivation surfaces and a sinuous crest-line of the unit bar. This resulted in curved cross strata in across-stream views. The geometry of the crest-line and lee side of the unit bar thus controls the shape of strata viewed normal to the flow direction.

4.4. Flume experiments with solitary unit bars in sandy gravel

Solitary unit bars in sandy gravel were 0.1 to 0.15 m high, and water depth over the trough and back were 0.2 m and 0.05 to 0.1 m, respectively. Bedforms superimposed on unit bars were dunes and bedload sheets that were 0.3 to 0.6 m long and up to 0.02 m high (Fig. 18A). The cross strata are on the order of tens of millimeters thick (up to 40 mm), and their thickness and grain size vary considerably (Fig. 18C). Coarsest grains accumulate at base of the cross-set, and rest sharply on bottomsets made of fine-grained sand (Fig. 18B). Relatively thick cross strata formed of coarse-grained open-framework gravel are due to passage of the highest superimposed bedforms, whereas thinner cross strata composed of sandy gravel are due to the passage of lower superimposed bedforms. Grain flows in sandy gravels did not form cross strata, only grain size segregation (Fig. 10).

5. Discussion

5.1. Relationship between cross-sectional areas of cross strata and superimposed bedforms

A simple model for cross-stratal area and thickness associated with superimposed bedforms, as seen in a vertical cross-section parallel to flow direction, is:

$$A = H_s L_s / 2 = H \text{ th} / \sin \alpha \quad (1a)$$

$$\text{th} = H_s L_s \sin \alpha / 2H \quad (1b)$$

where A is cross-sectional area of the superimposed bedform (assuming triangular cross-section) and the cross stratum, H_s and L_s are the height and length of the superimposed bedforms respectively, H is the height of the host bedform, th is thickness of the cross stratum, and α is the slope of the lee side of the host bedform (residual angle-of-repose). Eqs. (1a), (1b) does not apply if the H_s/H is greater than about 0.2, because such relatively large superimposed bedforms would cause reduction in slope of the lee side of the host bedform. Eqs. (1a), (1b) also does not apply if the cross-sectional area of superimposed bedforms is less than that of the wedge of sediment that fails and becomes a grain flow. In this case, the presorting in the superimposed bedform would be lost in the accumulating wedge. If the cross-sectional area of the superimposed bedform is greater than that of the failing wedge, as in these experiments, it may be possible to recognize internal cross strata related to individual grain flows.

It is important to recognize that the cross-sectional area of cross strata in preserved sets of cross strata is much less than that in fully preserved bedforms, as considered in these flume experiments. According to the theory of [Leclair and Bridge \(2001\)](#), approximately two-thirds of the height of a bedform, on average, is removed by erosion of the upper part. However, in these experiments, only a small amount of the upper part of the host bedform and cross-set is removed by superimposed bedforms.

[Fig. 19](#) shows that there is close correspondence between the cross-sectional areas of superimposed bedforms and cross strata for the case of bedload sheets on dunes, and bedload sheets, ripples and dunes on unit bars. The data used ([Table 3](#)) include not only these experiments, but our field data from the South Saskatchewan River in Canada and the Congaree River in South Carolina (plus extra data from [Levey, 1978](#)).

In the case of fully developed dunes, it is likely that L_s and H are linearly related to flow depth, and H_s may

be related to flow depth and/or grain size. Therefore, cross-stratal thickness should be related to flow depth and/or grain size. In the case of bedload sheets or dunes superimposed on unit bars, it is possible that H_s , L_s and H are linearly related to depth, such that cross-stratal thickness would also be related to depth. These ideas will be tested with data that we are collecting. If a relationship can be developed between thickness of cross strata, thickness of cross-sets, the geometry of superimposed and host bedforms, and hydraulic parameters such as flow depth and mean flow velocity, it would be very useful for reconstructing flow conditions in ancient deposits.

5.2. Effects of long-term flow unsteadiness

It is commonly thought that long-term changes in water discharge, associated for example with weather-related floods or diurnal melting and freezing, can result in variation in the grain size of adjacent cross strata. The grain size is normally assumed to increase as discharge increases. However, preliminary experiments show that this is a rather simplistic view. For example, if flow velocity and bed shear stress are increased in time, the top of a bar may be eroded if the increased sediment transport rate exceeds the upstream supply. If the finest-grained sediment occurs at the top of the bar deposit (coarsest at the base), erosion of the bar top will cause a decrease in grain size in sediment delivered to the lee side of the bar ([Kleinbans, 2002](#); [Lunt and Bridge, 2007](#)). If erosion proceeds, the sediment delivered to the

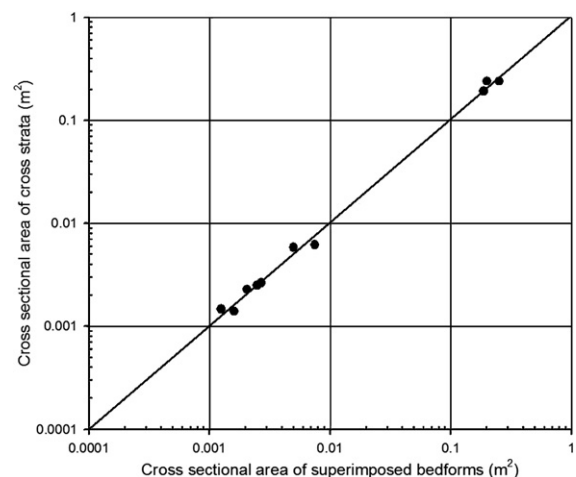


Fig. 19. Comparison of cross-sectional areas (along-stream view) of cross strata and superimposed bedforms for bedload sheets superimposed on sand dunes, and ripples, bedload sheets, and dunes superimposed on sandy and gravelly unit bars. Data sources shown in [Table 3](#).

Table 3
Data sources for Fig. 19

Source of data	Ripple area (m ²)	Strata area (m ²)	Host bedform type	superimposed bedform type
Experiment 1	0.0050	0.0058	Unit bar	Ripples
Experiment 2	0.0027	0.0027	Unit bar	Ripples
Experiment 3	0.0013	0.0015	Unit bar	Ripples
Experiment 4	0.0020	0.0023	Unit bar	Ripples
Experiment 5	0.0025	0.0025	Dune	Bedload sheets
Sandy gravel experiments 2005	0.0075	0.0062	Unit bar	Dunes, bedload sheets
Congaree 2006	0.1875	0.1920	Unit bar	Dunes
Congaree 2004	0.2000	0.2400	Unit bar	Dunes
South Saskatchewan 2006	0.0016	0.0014	Unit bar	Ripples
Levey (1978)	0.2520	0.2400	Unit bar	Dunes

lee side of the bar will become increasingly coarse. Another, related way of changing the sediment supply to the front of a bar is for a bank-slump to deliver a large slug of sediment into the river, possibly resulting in a migrating unit bar. The finest grain fractions outrun the coarser fractions, and this fine material can be deposited on the lee side of previously existing unit bars before the coarser sediment arrived. Therefore, a fine-coarse couplet in unit bar cross strata does not necessarily mean changing flow stage.

5.3. Further research and its implications

The full implications of linking the formation of cross strata with superimposed bedforms and longer-term flow unsteadiness will only be realized after more detailed experimental studies. It is first necessary to undertake comprehensive studies of the 3D geometry and grain size sorting of superimposed bedforms, the relationship between the passage of superimposed bedforms and the nature of grain flow on the lee side of the host bedform, and how these relate to the 3D geometry and grain size sorting of cross strata. The next step is relating the geometry, migration rate, and grain size sorting of both the host bedforms and the superimposed bedforms to flow and sediment transport conditions. Then the geometry and grain size sorting of cross strata can also be related to flow and sediment transport conditions. However, the preservation ratio (host bedform height/cross-set thickness) of cross strata must be known in order to do this (Leclair and Bridge, 2001).

Textural variations in cross strata can be related to spatial variations in porosity and permeability (reviews in Brayshaw et al., 1996; Tidwell and Wilson, 2000; Bridge, 2003). The improved descriptions and interpretations of cross strata associated with dunes and unit

bars proposed here should lead to much more reliable predictions of spatial variation in porosity and permeability, and thus to characterization of aquifers and hydrocarbon reservoirs. Numerical models that predict the effect of sedimentary structures on porosity and permeability (e.g. Kortekaas, 1985; Kelkar and Gupta, 1988; Ringrose et al., 1992; Scheibe and Freyberg, 1995; Scheibe and Murray, 1998) require realistic representation of the geometry and grain size sorting of different scales of cross strata. This is yet to be attained.

6. Conclusions

Grain size sorting in cross strata formed by dunes and unit bars is controlled by presorting of sediment supplied to their lee sides as well as by sorting by grain fall and grain flow on their lee sides. Presorting of sediment is associated with superimposed bedforms (e.g., bedload sheets, ripples, small dunes) and with longer-term flow unsteadiness. Variations in the size and grain size sorting of superimposed bedforms result in variations in the thickness and grain size of the cross strata. The thickest and coarsest cross strata are associated with the largest superimposed bedforms. Long-term flow unsteadiness is related to factors such as cyclic floods, associated with tides and melting-freezing cycles, erosion and deposition of upstream bars, and upstream bank-slumps, each of which can give rise to alternations of coarse- and fine-grained cross strata.

Detailed experimental studies are required to fully explore the relationship between presorting of sediment arriving at the lee sides of dunes and unit bars, and variations in the geometry and grain size sorting of their cross strata. Such studies will allow much more refined quantitative interpretation of ancient flow and sediment transport conditions (e.g., flow depth and velocity) from cross strata, and more accurate prediction of spatial variation in porosity and permeability, a major concern to hydrogeologists and petroleum geologists.

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